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Professional Summary

With overall 8+ of experience in the IT industry, I specialize in Big Data, Data Engineering primarily focusing on the design, optimization, and management of large-scale data pipelines with proven track record of enhancing data processing efficiency by ensuring high throughput and low latency, while consistently meeting business requirements by tackling complex data integration challenges and delivering comprehensive end-to-end data solutions, encompassing data governance, seamless integration, and high-performance processing using the services like **PySpark, Airflow, Databricks, Snowflake** and the cloud services such as **Aws** and **Azure**.

- Expert in **Hadoop, Spark**, for building scalable and fault-tolerant data processing frameworks. Skilled in developing and optimizing Spark-based ETL jobs for large datasets, including tuning performance and managing cluster resources for both real-time and batch processing solutions.
- Expertise in Utilizing **Databricks** for building scalable and efficient data processing pipelines, utilizing its integration with **Apache Spark** to optimize performance and manage large datasets effectively. Skilled in developing end-to-end machine learning models and streamlining workflows through Databricks' unified analytics environment.
- Proficient in utilizing **Snowflake's** cloud data platform to design and implement scalable, high-performance data architectures.
- Extensive experience with **AWS** services like **EMR, EC2, S3, Lambda, Glue, RDS, Redshift, DynamoDB, Sage Maker, Kinesis, CloudWatch, Athena, Aurora, Step Functions, SNS, SQS, CloudFormation, ECS, EKS, and Elastic Beanstalk**. Skilled in designing and optimizing scalable, high-performance cloud-based data pipelines.
- Experienced with **Azure** services like **Blob Storage, Data Lake Storage, Data Factory, Synapse Analytics, SQL Database, Databricks, Kubernetes Service, Event Hub, Cosmos DB, and Log Analytics**. Skilled in building scalable, secure data pipelines using **Terraform** and **ARM templates**, with secure secret management via **Key Vault** and proactive monitoring using **Azure Monitor**.
- Deep expertise in **SQL** databases like **Oracle, MySQL, PostgreSQL, DB2** for data modeling, querying, and optimization. Proficient in **NoSQL** tools like **Hive, Cassandra, and MongoDB** for handling large-scale, unstructured data and real-time processing.
- Experienced in **CI/CD pipelines** using **Jenkins, Docker, and Kubernetes** to automate and scale microservices-based data pipelines. Proficient in version control with **Git, GitLab, GitHub, and SVN**, ensuring code stability and efficient delivery through the **SDLC** process.
- Experience with **machine learning models** using **Scikit-Learn** and **TensorFlow** for predictive analysis and anomaly detection. Integrated ML models into data pipelines.
- Proficient in **Apache Airflow**, an open-source platform for orchestrating and automating complex workflows and data pipelines to enable seamless scalability and effective error handling, making it essential for managing data-driven processes efficiently.
- Managed Hadoop clusters using **Cloudera** and **Hortonworks**, configuring and optimizing clusters for large-scale data processing. Oversaw data replication, security, and resource allocation.
- Experienced in **ETL** tools like **Informatica** and **Talend** for designing and implementing data extraction and transformation processes. Designed data models using **Erwin** to support analytical and operational reporting, ensuring data consistency and integrity.
- Skilled in handling diverse data formats including **CSV, Parquet, ORC, Avro, JSON, XML, and TXT**. Built data ingestion frameworks to standardize and transform structured and semi-structured data while maintaining data quality and integrity.
- Developed interactive dashboards using **Tableau, Power BI, and Looker** to enable data-driven decision-making. Created real-time visualizations to track KPIs and business metrics, ensuring seamless integration with data warehouses and streams.
- Demonstrated strong documentation skills using **JIRA, ServiceNow, Confluence** for collaborative project documentation, **SharePoint** for secure document management, and **Lucidchart** for detailed data flow diagrams.
- Highly motivated and proactive team player with strong problem-solving skills and a passion for learning new technologies. Excellent communication and leadership skills with a commitment to delivering high-quality solutions aligned with business objectives.

TECHNICAL SKILLS:

Big Data Tools	Apache Spark, Hive, Sqoop, Kafka, Flink, Snowflake
Cloud Technologies	AWS: EMR, EC2, S3, Lambda, Glue, RDS, Redshift, DynamoDB, Athena, Kinesis, CloudWatch, CloudTrail, Amazon SNS, Amazon SQS, Amazon Step Functions Azure: Blob Storage, Azure Data Lake Storage, Azure Data Factory, Synapse Analytics, Azure SQL Server, data bricks, Azure Kubernetes Service, Cosmos DB, Stream Analytics, Azure SQL data Warehouse, Azure event bridge,
Languages	Python, PySpark
Hadoop Distributions	Cloudera, Hortonworks
ETL TOOLS	Apache Nifi, Apache Airflow, Informatica, Talend, Google Cloud Factory, SQL Server Integration Services
Databases	SQL: Oracle, MySQL, PostgreSQL, DB2 NoSQL: Hive, Cassandra, MongoDB
BI and Visualization	Tableau, Power BI, Looker
Version Control Tools	Git, GitLab, GitHub, SVN
Methodologies	Software Development Life Cycle (SDLC) CI/CD Pipelines: Jenkins, Docker, Kubernetes Project Management: JIRA, ServiceNow, Rally, Azure Dashboard
Machine Learning	Scikit-Learn, TensorFlow

PROFESSIONAL EXPERIENCE:

Sr DATA ENGINEER
DXC Technology

Jan 2024 - present

Responsibilities:

- Designed secure, scalable **AWS** cloud architectures for real-time inventory tracking and multi-channel customer experiences, ensuring **PCI-DSS** compliance and cost optimization. Used **Terraform**, **JIRA**, and **CloudWatch** for automation, monitoring, and team collaboration.
- Collected retail data requirements to improve sales, inventory, and customer insights. Built ETL pipelines using **AWS Glue**, and **Talend**, integrating data from **POS** and **CRM** systems, and ensured smooth data flow with monitoring and documentation.
- Built ETL pipelines using **AWS Glue**, **S3**, **Redshift**, **Lambda**, **DynamoDB**, **RDS**, **EMR**, and **EC2** to integrate retail data from **POS** and **CRM** systems, driving insights into sales, inventory, and customer behavior.
- Set up monitoring and logging with **Amazon CloudWatch** and **AWS CloudTrail** for efficient data flow and accurate analytics, automating workflows to support retail operations.
- Utilized **Hadoop**, **PySpark**, **Hive**, **Snowflake**, and **Databricks** to design and optimize large-scale data pipelines, improving data integration, transformation, and querying efficiency.
- Designed data pipelines for **ingestion, integration, and aggregation** using **Kafka** for real-time streaming, processing data from **APIs** and migrating **Oracle** and **SQL Server** to **Amazon S3** for scalable analytics.
- Developed and executed **on-premises** to cloud migration strategies using **AWS Data Sync** and **Cloud Endure**, ensuring secure data transfer, optimized cloud infrastructure, and stable post-migration performance.
- Built OLAP cubes using **AAS** and **Kyvos**, enabling advanced data analysis and connected them to **Excel** and **BI tools** for interactive dashboard creation.
- Integrated **Amazon DynamoDB** into the retail project for high-performance, low-latency data storage, optimizing key-value and document-based data handling for **inventory tracking** and **customer behavior analysis** within the **ETL pipeline**.
- Automated ETL pipeline validation using **Pytest** and **unit test**, ensuring accurate data transformation and integration. Designed tests to detect anomalies and enforce business rule compliance.

- Implemented data quality checks using **AWS Glue Data Brew** and custom scripts to ensure consistency and integrity, applying profiling and error-handling measures to maintain high reliability standards.
- Implemented **Log4j** for logging and exception handling in ETL pipelines, configuring **INFO**, **DEBUG**, and **ERROR** log levels to track execution, capture errors, and ensure quick issue resolution for optimal system reliability and performance.
- Developed interactive dashboards and reports using **Power BI**, **Tableau**, and **Quick Sight**, while creating secure **APIs** for data sharing. Queried datasets with **Amazon Athena** and **Snowflake** and built **OLAP cubes** for advanced multidimensional analysis and visualization.
- Used **Parquet**, **ORC**, and **Avro** for optimized data storage and retrieval, reducing storage costs and improving query performance in large-scale analytics workflows, while ensuring compatibility with big data tools.
- Implemented **data encryption**, **row-level security**, and **column masking** to protect sensitive information while ensuring workflows complied with **GDPR** and **DPA** privacy policies.
- Enforced access control using **Apache Ranger**, **IAM Policies**, and **custom logic** to ensure secure, role-based, and compliant data handling.
- Used **Docker** for containerization and **Kubernetes** for scalable, fault-tolerant deployments with automated workflows, secure networking, and performance monitoring using tools like **Prometheus** and **Grafana**.
- Utilized advanced scheduling and Implemented tools such as **Apache Airflow** to efficiently automate workflows, optimize task management, and ensure seamless execution of complex data pipelines across systems.
- Streamlined CI/CD processes with tools like **Jenkins**, automating build, test, and deployment workflows. Ensured efficient and reliable application delivery across all environments.
- Used **GitHub** for version control to manage repositories and track code changes effectively. Enhanced collaboration and code quality through branching strategies and pull requests.
- Built and deployed machine learning models using **TensorFlow**, **PyTorch**, and **scikit-learn**, optimizing performance with preprocessing. Automated workflows with **Kubeflow**, deployed models via **MLflow** and **SageMaker**, and visualized insights using **Matplotlib** and **Seaborn**.
- Implemented **Agile SDLC** with sprint-based planning and iterative development. Utilized **Jira** for task tracking and collaboration

Environment: Glue, S3, Redshift, Lambda, DynamoDB, RDS, EMR, EC2, Terraform, JIRA, CloudWatch, AWS Data Sync, Cloud Endure, Amazon Athena, Hadoop, PySpark, Hive, Snowflake, Databricks, Kafka, AWS Glue Data Brew, Log4j, AAS, Kyvos, Power BI, Tableau, Quick Sight, Parquet, ORC, Avro, AWS IAM Policies, Apache Ranger, GDPR, DPA Privacy Policies, Data Encryption, Row-Level Security, Column Masking, Docker, Kubernetes, Prometheus, Grafana, Apache Airflow, Jenkins, GitHub, TensorFlow, PyTorch, scikit-learn, Kubeflow, MLflow, SageMaker, Matplotlib, Seaborn, Agile SDLC, Jira.

DATA ENGINEER

Betis Group Inc

Jan 2023 – dec 2023

Responsibilities:

- Designed a scalable and fault-tolerant data pipeline using **Azure** to process large volumes of telecom data, ensuring high availability, low latency, and efficient handling of real-time and batch processing.
- Developed scalable ETL workflows using **Azure Data Factory (ADF)** and **PySpark** to efficiently extract, transform, and load structured and unstructured telecom data from diverse sources, ensuring high data quality and performance.
- Integrated data from on-prem systems into **Azure Blob Storage** and processed it using **Azure Synapse**, **Azure Data Lake**, and **Azure SQL** to enhance data accessibility and processing efficiency. Utilized **Azure Data Factory** for automated data ingestion and transformation.
- Deployed real-time data streaming using **Azure Event Hubs** and **Azure Stream Analytics**, enabling low-latency data processing and improving network performance insights.
- Enhanced data processing using **Azure Synapse** to create scalable, high-performance analytical models and improve query performance for large datasets.
- Utilized **Azure Data Factory** to Developed data movement and transformation across various sources, ensuring streamlined data integration and consistency.
- Developed advanced data transformation and machine learning models using **Azure Databricks**, enabling efficient processing of complex datasets and improving analytical insights.
- Processed large-scale network logs and customer data using **Hadoop** and **Spark-Scala** for fast data aggregation and real-time analysis.

- Migrated large telecom datasets from **Teradata** to **Snowflake** using both batch and real-time processing methods, Utilizing Snowflake's native data migration tools.
- Developed multi-dimensional models and built analytical cubes using **Azure Analysis Services (AAS)** and **Kyvos** to optimize data exploration and enable fast, interactive analysis, enhancing decision-making for telecom business operations.
- I optimized data consumption by designing efficient access patterns using **Snowflake** queries, leveraging and enabling direct access to data through BI tools like **Power BI** and **Tableau** for seamless reporting and analysis.
- Implemented encryption, **row-level security**, and **column masking** to safeguard sensitive data, ensuring compliance with **GDPR** and **DPA** policies to maintain privacy and security across the system.
- Figured access control using **Ranger** and **IAM policies** to enforce secure, role-based data access, ensuring that users only had appropriate permissions to access sensitive data.
- Deployed data processing jobs using **Docker** and **Kubernetes**, enabling scalability and faster deployment cycles, which streamlined the management and Execution of data workflows.
- Automated workflow execution using **Airflow** and **Control-M**, ensuring seamless orchestration and efficient operation of data pipelines. This streamlined scheduling and improved the reliability of data workflows.
- Used **Bitbucket** to manage the codebase and track changes in data pipelines, facilitating seamless team collaboration and ensuring version control for efficient development
- Using **Azure DevOps** to automate CI/CD pipelines, enabling faster deployment, improving build quality, and enhancing development efficiency.

Environment: Azure Data Factory (ADF), PySpark, Azure Synapse, Hadoop, Spark-Scala, Snowflake, Azure Analysis Services (AAS), Kyvos, Hive, Power BI, Parquet, Encryption Tools, Ranger, IAM Policies, Docker, Kubernetes, Airflow, Control-M, Bitbucket, Azure DevOps, Azure Synapse, Azure Data Factory, Azure Databricks.

Kaiser Permanente **Data Analyst**

Jan 2020 – Sep 2022

Responsibilities:

- Consolidated electronic health records (EHR) and insurance data into centralized data lakes using **Informatica** and stored them in **Hadoop HDFS** for scalable processing.
- Processed large volumes of healthcare claims data with **Spark** to identify trends and ensure accurate reimbursements while reducing processing delays.
- Built analytical queries with **Hive** to gain insights into patient admission patterns, treatment outcomes, and resource allocation.
- Designed robust ETL workflows with **Informatica** for extracting, transforming, and loading clinical trial and prescription data into **Hadoop** systems.
- Used **Spark Streaming** for real-time healthcare transaction processing, enabling faster eligibility checks and claims submissions.
- Scheduled batch processing jobs with **Oozie**, streamlining periodic reporting and compliance data workflows across the Hadoop ecosystem.
- Automated data cleansing and validation using **Informatica** to ensure the reliability and consistency of healthcare datasets.
- Analyzed population health metrics using **Hive**, **Pig**, and **Hadoop**, identifying trends for proactive healthcare interventions.
- Deployed distributed storage systems with **Hadoop HDFS** to efficiently store unstructured healthcare datasets with high scalability.
- Developed predictive models on **Spark** for readmission risk predictions and disease outbreak forecasting, enhancing patient care strategies.
- Automated regulatory compliance report generation using **Hive** to meet standards like **HIPAA** while maintaining data integrity.
- Processed large-scale clinical trial data using **Pig**, enabling researchers to derive insight into evidence-based medicine.
- Aggregated diverse datasets, including lab results and pharmacy data, with **Hive** and **Spark**, creating unified healthcare records.

- Orchestrated complex ETL workflows with **Oozie**, ensuring accurate and timely processing of healthcare operational data.
- Enforced strong encryption and access controls using **Informatica** and **Hadoop security** to protect sensitive healthcare information.
- Used **Pig** scripts to detect fraudulent activities by analyzing historical claim patterns and identifying anomalies.
- Processed data adhering to interoperability standards like **HL7** and **FHIR** with **Hive** and **Spark**, enabling seamless system integration.
- Transformed raw clinical data into analytical formats with **Informatica** and stored them in **Hive** tables for efficient querying.
- Improved query performance by leveraging **Spark SQL** and optimizing partitioned **Hive tables** for faster data retrieval.
- Scheduled workflows with **Oozie** for time-sensitive healthcare data processing, ensuring critical deadlines were met seamlessly.

Environment: Informatica, Hadoop HDFS, Spark, Hive, Spark Streaming, Oozie, Pig, Hadoop security, Spark SQL, Hive tables, FHIR, HIPAA.

Python Developer Bank of Montreal

Sep 2017 – Jan 2020

Responsibilities:

- Build a centralized financial data system by handling ETL processes with **SQL** and performing data cleaning and transformation using **Python's Pandas**.
- Predict stock prices by training machine learning models with **Python's Scikit-learn** and accessing historical stock data via **SQL queries**.
- Create a personal finance tracker, using **Python** for automation and interaction, and **SQL** to store and categorize expense data.
- Detect fraud by setting **SQL triggers** for suspicious activities and developing predictive models with **Python's Scikit-learn**.
- Develop an interactive budget planner with **Python's Dash** for visualization and **SQL** to manage budget and transaction data.
- Forecast future revenues by building analytics models with **Python's TensorFlow** and feeding them historical data from **SQL**.
- Predict credit scores with machine learning models built using **Python** and utilize **SQL** to fetch user financial datasets.
- Analyze cash inflows and outflows, using **Python** for calculations and reports, and **SQL** for data storage and retrieval.
- Automate tax computation with **Python's NumPy** for numerical operations, storing tax rules and brackets in **SQL tables**.
- Build a loan eligibility tool by training decision-making models with **Python's Scikit-learn** and storing profiles in **SQL databases**.
- Monitor financial KPIs dynamically through real-time dashboards built with **Python's Matplotlib**, retrieving data using **SQL**.
- Analyze portfolio diversification with **Python's financial libraries**, retrieving asset allocation data stored in **SQL**.
- Automate profit and loss calculations using **Python** and query financial data for statements through **SQL databases**.
- Track cryptocurrency trends by fetching live prices using **Python APIs** and storing historical price data in **SQL** for comparison.
- Generate financial reports automatically using **Python's Report Lab**, retrieving and formatting data directly from **SQL databases**.

Environment: Python, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, Dash, Matplotlib, APIs (requests), BeautifulSoup, ReportLab, NLTK, SpaCy, SQL queries, ETL processes, SQL joins, SQL triggers, SQL tables.

Education:

Campbellsville University, DEC 2024

Master of Science in computer science, GPA 3.8